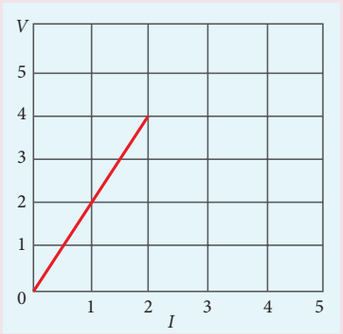
I Multiple Choice Questions

1. The following graph shows current versus voltage values of some unknown conductor. What is the resistance of this conductor?

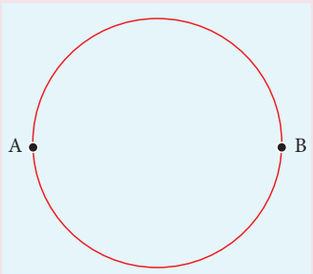


O20LT

(a) 2 ohm (b) 4 ohm

(c) 8 ohm (d)1 ohm

2. A wire of resistance 2 ohms per meter is bent to form a circle of radius 1m. The equivalent resistance between its two diametrically opposite points, A and B as shown in the figure is



(a) $\pi\ \Omega$ (c) $\frac{\pi}{2}\ \Omega$

(c) $2\pi\ \Omega$ (d) $\frac{\pi}{4}\ \Omega$

3. A toaster operating at 240 V has a resistance of 120 Ω. Its power is

a) 400 W b) 2 W

c) 480 W d) 240 W

4. A carbon resistor of $(47 \pm 4.7)\ k\Omega$ to be marked with rings of different colours for its identification. The colour code sequence will be

a) Yellow – Green – Violet – Gold

b) Yellow – Violet – Orange – Silver

c) Violet – Yellow – Orange – Silver

d) Green – Orange – Violet - Gold

5. What is the value of resistance of the following resistor?



(a)100 k Ω (b)10 k Ω

(c) 1k Ω (d)1000 k Ω

6. Two wires of A and B with circular cross section are made up of the same material with equal lengths. Suppose $R_A = 3\ R_B$, then what is the ratio of radius of wire A to that of B?

(a) 3 (b) $\sqrt{3}$

(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{3}$

7. A wire connected to a power supply of 230 V has power dissipation P_1. Suppose the wire is cut into two equal pieces and connected parallel to the same power supply. In this case power dissipation is P_2. The ratio $\frac{P_2}{P_1}$ is

(a)1 (b) 2

(c) 3 (d) 4

12th Physics_Vol-1_Unit 2_English Medium.indd 118 12-12-2022 17:26:13